

Mattéo Couplet

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EDUCATION

UCLouvain (Belgium)

PhD in Applied Mathematics

Sep. 2024

MSc in Applied Mathematics (Summa cum laude)

June 2020

BSc in Mechanical Engineering (Summa cum laude)

June 2018

EXPERIENCE

Boston University

Oct. 2024 – Present

Postdoctoral Associate – BAEF and WBI Fellow – Group of Prof. Edward Chien

Boston, MA

- **Built an end-to-end quad meshing pipeline** with controllable element sizing, soon part of Gmsh. Designed for simulation robustness and performance on industrial CAD models.
- **Developed geometry algorithms for manufacturing**, notably a curl quantization method that automatically inserts stitch irregularities in knit patterns for automatic computational knitting.
- Led research directions through grant writing, mentored PhD researchers, and delivered research code used across the lab.

UCLouvain

Oct. 2020 – Sep. 2024

PhD Researcher – FNRS Fellow – Advised by Prof. Jean-François Remacle

Louvain-la-Neuve, Belgium

- **Designed a new framework for structured quad and hex meshing** based on integrable frame fields, improving robustness and control for industrial geometries.
- **Created a CAD-aware block-structured meshing pipeline** using integer optimization to enforce topology constraints and achieve high-order solver readiness.
- Teaching roles in FEM, numerical analysis, and computational geometry.

MIT

June – Aug. 2019

Research Intern – Lhoist Berghmans Fellow – Advised by Prof. Laurent Demanet

Cambridge, MA

- Implemented a method for porous media reconstruction using deep generative networks. The model builds a statistical 3D representation of a porous medium from ground truth 2D sections.

Freemind Consulting

June – Aug. 2018

R&D Intern

Brussels, Belgium

- Developed, tested and documented a new algorithm that leverages the latest signal processing techniques for fingerprinting and watermarking. The code is now part of a solution that guarantees ownership data of sound recordings.

AWARDS

- **Belgian American Educational Foundation and Francqui Foundation Research Fellowship (2024).** Competitive one year postdoctoral grant (\$50k) awarded by leading philanthropy organizations.
- **Wallonia-Brussels International Excellence Fellowship (2024).** Competitive two year postdoctoral grant (\$58k) awarded by a regional government of Belgium.
- **FNRS Research Fellowship (2020).** Four year PhD grant (\$136k) awarded to top-ranking students by the national science foundation of Belgium.
- **Lhoist Berghmans Innovation Chair MIT Scholarship (2019).** Awarded to top-ranking students to fund a summer research internship at MIT.
- **Competitive Programming.** First place at the Belgian national contest (2015 – participated in IOI), First place

TEACHING

- Numerical Analysis, UCLouvain, Fall '20, Spring '22, Spring '23, Spring '24
- Advanced Numerical Methods, UCLouvain, Spring '22, Spring '23
- Computational Geometry, UCLouvain, Fall '22, Fall '23
- Finite Element Method, UCLouvain, Spring '21, Spring '22

ADVISING/MENTORING

- Daniel Smolders, Master's Student at UCLouvain, 2025 – present
- Rahul Mitra, Doctoral Student at Boston University, 2024 – 2026
- Loïs Dermine, Master's Student at UCLouvain, 2023 – 2024

REVIEWING

- Engineering with Computers (1 technical paper)
- ACM Symposium on Computational Fabrication 2026 (1 technical paper)
- ACM SIGGRAPH 2026 (3 technical papers)
- ACM SIGGRAPH 2025 (3 technical papers)
- SIAM International Meshing Roundtable 2025 (1 technical paper)

PUBLICATIONS

- [Couplet, M.](#) & Mitra, R. (equal contrib's), Liu, R., Ng, J., Markov, D., Samosir, W., Hofmann, M., & Chien, E. (2026). Surface Power Diagrams for Knit Singularity Placement. In *ACM Transactions on Graphics*. [DOI](#).
- Pacheco-Tallaj, N., [Couplet, M.](#), Chien, E., & Palmer, D. (2026). Synchronizing Fields with Singularities. In *Proceedings of SIGGRAPH 2026 Conference Papers*. [DOI](#).
- [Couplet, M.](#), Chemin, A., Bommès, D., & Chien, E. (2026). Surface Quadrilateral Meshing from Integrable Odeco Fields. Presented at the 2026 *Symposium on Geometry Processing*, to appear in *Computer Graphics Forum*. [arXiv](#).
- [Couplet, M.](#), Chemin, A., & Remacle, J.-F. (2026). Size-Controlled Quadrilateral Meshing using Integrable Odeco Fields. In *Computer-Aided Design*. [DOI](#).
- Mitra, R., [Couplet, M.](#), Wang, T., Hoffman, M., Wu, K., & Chien, E. (2025). Curl Quantization for Automatic Placement of Knit Singularities. In *Proceedings of SIGGRAPH 2025 Conference Papers*. [DOI](#).
- [Couplet, M.](#), Chemin, A., & Remacle, J.-F. (2024). Integrable Frame Fields using Odeco Tensors. In *Proceedings of the 2024 SIAM International Meshing Roundtable*. [DOI](#).
- [Couplet, M.](#), Reberol, M., & Remacle, J.-F. (2021). Generation of High-Order Coarse Quad Meshes on CAD Models via Integer Linear Programming. In *ALAA Aviation 2021 Forum*. [DOI](#).

SOFTWARE

- [PowerKnit](#): Compute knit graphs for whole-garment machine knitting from a surface mesh or a sewing pattern. Implements the surface power diagrams method for placing knit singularities.
- [QMP](#): An end-to-end quadrilateral mesher for planar geometries based on integrable odeco fields, implementing the size-controlled quad meshing method.